Introduction to Database Management System (DBMS)

Presented by:
K.VIJAYALAKSHMI
192511189
B.E CSE

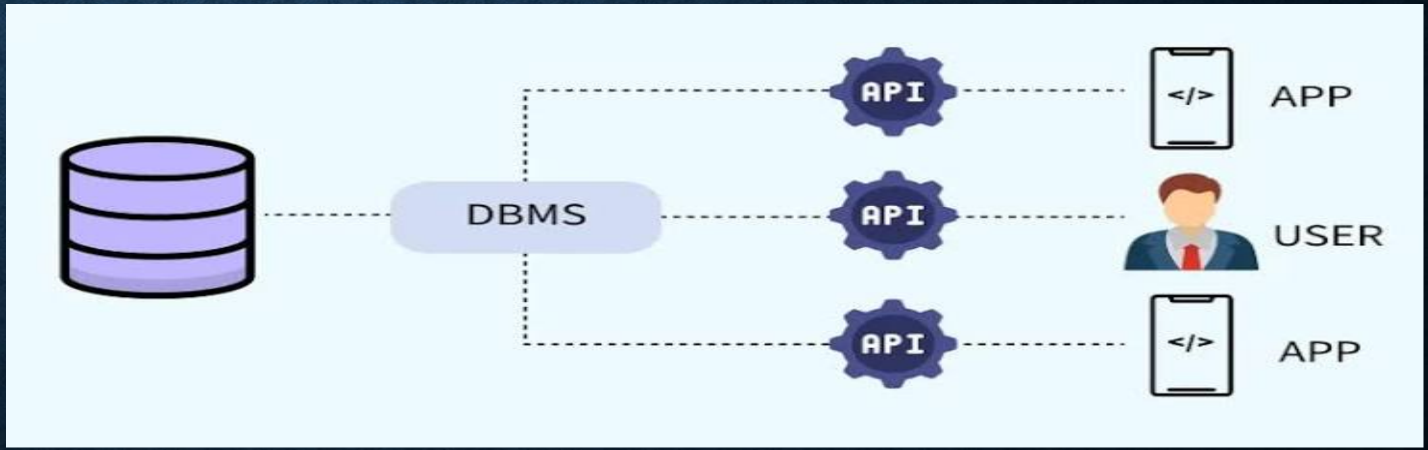
DEFINITION OF DBMS

What is DBMS?

- A **Database Management System (DBMS)** is software used to create, store, retrieve, and manage data efficiently.
- It acts as an interface between users and databases.
- It helps organize data in a structured and secure manner.

Examples:

1. MySQL
2. Oracle
3. Microsoft SQL Server

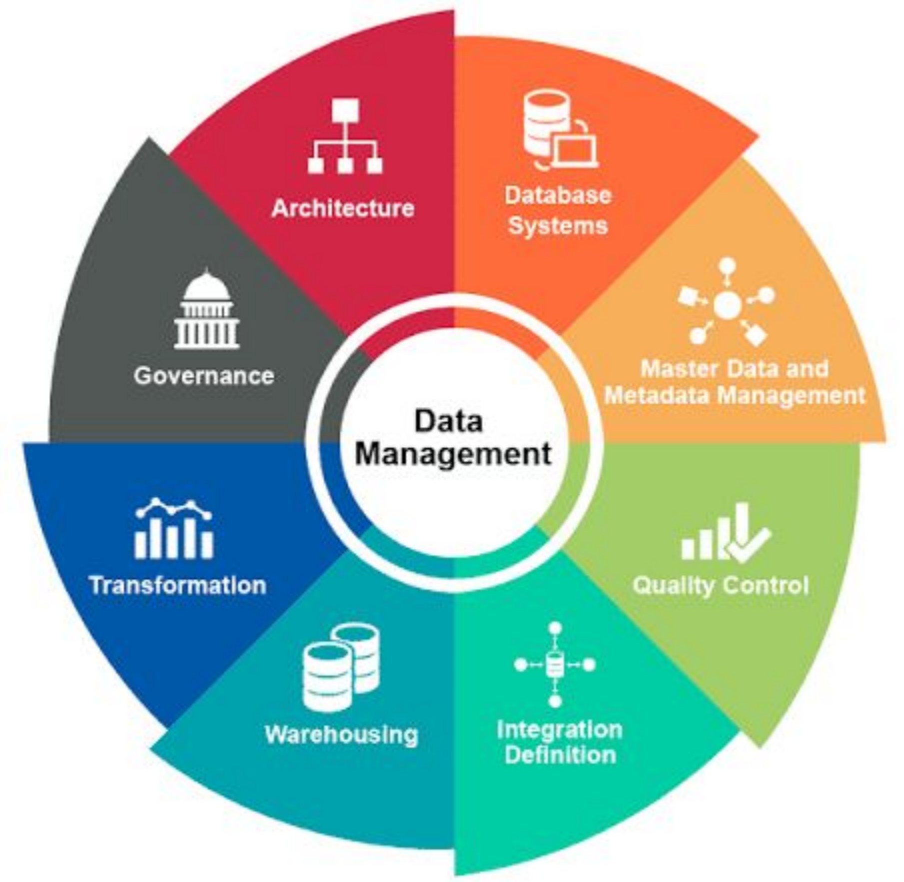


OBJECTIVES OF DBMS

- Store data efficiently.
- Reduce data redundancy (duplicate data).
- Ensure data security.
- Maintain data integrity and accuracy.
- Allow multiple users to access data simultaneously.
- Provide backup and recovery of data.

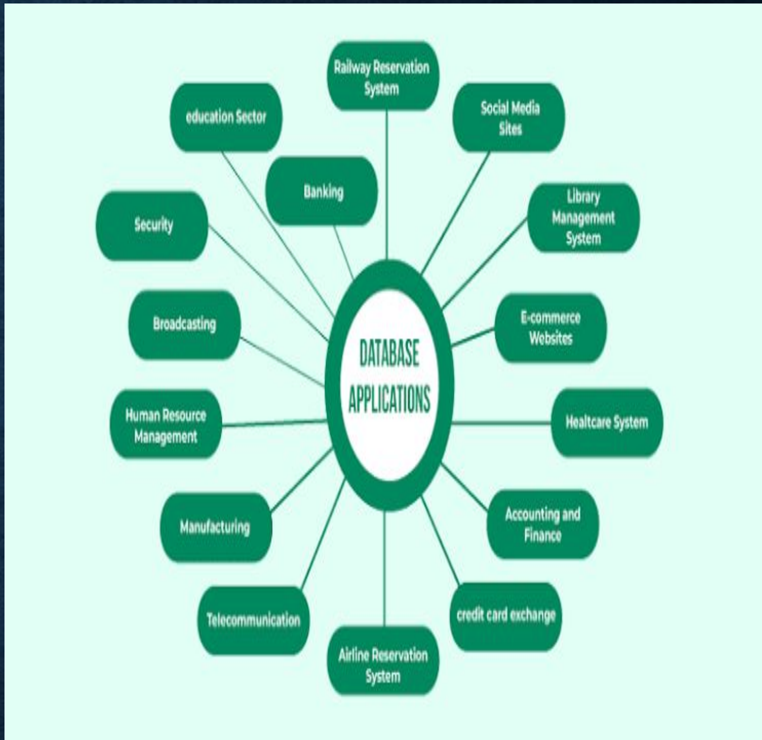
ADVANTAGES OF DBMS

- ❑ Easy data storage and retrieval.
- ❑ Improved data security.
- ❑ Reduced data duplication.
- ❑ Faster data access.
- ❑ Supports data sharing among users.
- ❑ Provides backup and recovery.
- ❑ Ensures data consistency.



REAL-WORLD APPLICATIONS

- **Banking:** Customer accounts and transactions.
- **Hospitals:** Patient records and appointments.
- **Education:** Student information and results.
- **E-commerce:** Orders, products, and customers.
- **Airlines/Railways:** Ticket booking and schedules.
- **Libraries:** Book catalog and member records.



Conclusion

- DBMS is software used to store, organize, and manage data efficiently.
- It reduces data redundancy and improves data security and accuracy.
- It enables fast data retrieval and supports multiple users simultaneously.
- DBMS is widely used in banking, healthcare, education, e-commerce, and many other industries.
- It plays a vital role in managing large amounts of data effectively.

THANK YOU